



Sir Syed University of Engineering & Technology, Karachi

Software Engineering (CS320)

BOOKS :

TEXTBOOK:

Software Engineering, Ian Sommerville, 10th Edition

REFERENCE BOOK:

Software Engineering: A Practitioner's Approach by Roger S Pressman , 8th Edition

HANDOUTS :

For some topics , handouts will be provided prior to the conduction of lectures

Course Assessment Methods:

Assessment Tools	Percentage (%)
Quizzes/Assignments	10 %
Lab Work	20%
Midterm Exam	20%
Final Exam	50%
Total	100 Marks

Chapter # 1 : Introduction

Topics covered

- Professional software development
- What is meant by software engineering.
- Software engineering ethics
- A brief introduction to ethical issues that affect software engineering.

Software Engineering

- The economies of ALL developed nations are dependent on software.
- More and more systems are software controlled
- Software engineering is concerned with theories, methods and tools for professional software development.

What is Software ?

Software can define as:

- Instruction – executed provide desire features, function & performance.
- Data structure – to adequately manipulate operation.
- Documents – operation and use of the program.
- Software products may be
 - **Generic** - developed to be sold to a range of different customers e.g. PC software such as Excel or Word.
 - **Bespoke (custom)** - developed for a single customer according to their specification.

Hardware vs. Software

Hardware

- Manufactured
- wear out
- Built using components
- Relatively simple

Software

- Developed/engineered
- deteriorate
- Custom built
- Complex

What is the difference between software engineering and computer science?

- Computer science is concerned with theory and fundamentals; software engineering is concerned with the practicalities of developing and delivering useful software.
- Computer science theories are still insufficient to act as a complete foundation for software engineering

What is the difference between software engineering and system engineering?

- System engineering is concerned with all aspects of computer-based systems development including hardware, software and process engineering. Software engineering is part of this process concerned with developing the software infrastructure, control, applications and databases in the system.
- System engineers are involved in system specification, architectural design, integration and deployment.

What is a software process?

- A set of activities whose goal is the development or evolution of software.
- Generic activities in all software processes are:
 - Specification - what the system should do and its development constraints
 - Development - production of the software system
 - Validation - checking that the software is what the customer wants
 - Evolution - changing the software in response to changing demands.

What is a software process model?

- A simplified representation of a software process, presented from a specific perspective.
- Examples of process perspectives are
 - Workflow perspective - sequence of activities;
 - Data-flow perspective - information flow;
 - Role/action perspective - who does what.

What are the costs of software engineering?

- Roughly 60% of costs are development costs, 40% are testing costs. For custom software, evolution costs often exceed development costs.
- Costs vary depending on the type of system being developed and the requirements of system attributes such as performance and system reliability.
- Distribution of costs depends on the development model that is used.

Software characteristics

- Software is developed or engineered; it is not manufactured.
- Software does not “wear out” but it does deteriorate.
- Software continues to be custom built, as industry is moving toward component based construction.

Component Based vs. Custom Built

- Hardware products typically employ many standardized design components.
- Most software continues to be custom built.
- The software industry does seem to be moving (slowly) toward component-based construction.

Changing nature of software

- System software
- Application software
- Embedded software
- Web applications
- Artificial intelligence software

Changing nature of software

- **System software** is a collection of programs written to service other programs.
- Some system software (e.g., compilers, editors, and file management utilities) process complex, but determinate, information structures.
- Other systems applications (e.g., operating system components, drivers, telecommunications processors) process largely indeterminate data

Changing nature of software

- **Real-time Software** monitors/analyzes/controls real-world events as they occur is called real time
- Include a data gathering component that collects and formats information from an external environment

Changing nature of software

Engineering and scientific software

- Characterized by "number crunching" algorithms.
- Applications range from astronomy to volcano logy, from automotive stress analysis to space shuttle orbital dynamics, and from molecular biology to automated manufacturing.

Ex. Computer Aided Design (CAD), system stimulation etc.

Changing nature of software

Embedded Software

- It resides in read-only memory and is used to control products and systems
- Embedded software can perform limited and esoteric functions.
- **Ex.** keypad control for a microwave oven.

Changing nature of software

Web Applications

- Web apps can be little more than a set of linked hypertext files.
- It evolving into sophisticated computing environments that not only provide standalone features, functions but also integrated with corporate database and business applications.

Changing nature of software

Artificial Intelligence software

- AI software makes use of non-numerical algorithms to solve complex problems that are not amenable to computation or straightforward analysis

Ex. Robotics, expert system, game playing, etc.

Software engineering ethics

- Software engineering involves wider responsibilities than simply the application of technical skills.
- Software engineers must behave in an honest and ethically responsible way if they are to be respected as professionals.
- Ethical behaviour is more than simply upholding the law.

Issues of professional responsibility

- Confidentiality

Engineers should normally respect the confidentiality of their employers or clients irrespective of whether or not a formal confidentiality agreement has been signed.

- Competence

Engineers should not misrepresent their level of competence. They should not knowingly accept work which is out with their competence.

Issues of professional responsibility

- Intellectual property rights

Engineers should be aware of local laws governing the use of intellectual property such as patents, copyright, etc.

- Computer misuse

Software engineers should not use their technical skills to misuse other people's computers. Computer misuse ranges from relatively trivial (game playing on an employer's machine, say) to extremely serious (dissemination of viruses).

ACM/IEEE Code of Ethics

- The professional societies in the US have cooperated to produce a code of ethical practice.
- Members of these organizations sign up to the code of practice when they join.
- The Code contains eight Principles related to the behavior of and decisions made by professional software engineers, including practitioners, educators, managers, supervisors and policy makers, as well as trainees and students of the profession.

Ethical principles

1. **PUBLIC** - Software engineers shall act consistently with the public interest.
2. **CLIENT AND EMPLOYER** - Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.
3. **PRODUCT** - Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
4. **JUDGMENT** - Software engineers shall maintain integrity and independence in their professional judgment.

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Ethical principles

5. **MANAGEMENT** - Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.

6. **PROFESSION** - Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.

7. **COLLEAGUES** - Software engineers shall be fair to and supportive of their colleagues.

8. **SELF** - Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession